

A 2D SIMULATION-BASED DECISION SUPPORT SYSTEM FOR FLOOD EVACUATION PLANNING

CAPSTONE PROJECT

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ABSTRACT

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1 Introduction

Flooding can disrupt road access, increase evacuation demand, and constrain the use of evacuation centers. During a flood event, disaster planners and responders must determine which routes remain feasible, which centers can accommodate affected residents, and how available vehicles and other resources can be distributed. These decisions become more difficult when road conditions, shelter capacity, affected population, and resource availability change across possible flood situations.

Maps and geographic information can show locations, roads, and hazard areas, but static representations alone provide limited support for testing how changing conditions affect evacuation options. A scenario-based decision-support system can integrate relevant geographic and operational information so that planners can configure conditions, examine their consequences, and compare alternative evacuation plans before an event occurs.

This study develops *OverFlow*, a geospatial decision-support system for flood-evacuation planning. The system is intended to help authorized local disaster planners and responders configure and compare scenarios involving rainfall, affected population, road or bridge closures, evacuation-center status and capacity, and available resources. It provides decision-support information on flood-affected areas, feasible and alternative evacuation routes, center allocation, estimated evacuation time, and resource allocation. The system supplements, rather than replaces, the judgment and authority of disaster-management personnel.

1.1 Project Context

Flooding can change the conditions under which an evacuation plan must operate. Roads and bridges may become unavailable, the number of people requiring assistance may vary

between areas, evacuation centers may have limited capacity, and available vehicles and other resources may be insufficient for every possible situation. Local disaster planners must consider these conditions together when determining which routes remain usable, where affected residents can be accommodated, and what alternatives are available when an expected route or center cannot be used.

Maps can identify roads, boundaries, waterways, facilities, and hazard-related locations, but a map by itself does not show the consequences of changing several planning conditions at once. An evacuation plan may need to be reconsidered when rainfall conditions change, a road is manually closed, an evacuation center becomes unavailable or reaches capacity, the affected population increases, or fewer resources are available. Examining these conditions as configurable scenarios gives planners a structured way to compare possible evacuation arrangements before selecting or revising a plan.

This project develops *OverFlow*, a localized two-dimensional simulation-based decision-support system for flood evacuation planning. The system is intended for authorized local disaster planners and responders. It combines terrain, road, boundary, evacuation-center, population, and resource information in an interactive planning environment. Users configure rainfall intensity and duration, affected population, road or bridge closures, evacuation-center status and capacity, and available resources. The system then presents flood-affected conditions and analyzes feasible and alternative routes, center allocation, estimated evacuation time, and resource distribution for comparison with other scenarios.

The flood component uses a digital elevation model as its terrain input. D8 flow direction and flow accumulation represent how terrain directs and concentrates water, while a simplified rainfall overlay and threshold classify areas for scenario analysis. Drainage infrastructure may be displayed as a reference layer but is not included in the flood compu-

tation. The rainfall-to-indicator formula, classification threshold, and validation procedure still require final definition; therefore, the output must be treated as a relative scenario representation rather than a calibrated hydraulic result or exact flood forecast.

OverFlow is intended to supplement the planning process by organizing relevant conditions and showing how changes in those conditions affect evacuation options. It does not make autonomous emergency decisions, guarantee that a route will remain safe during an actual event, or replace the authority and judgment of disaster-management personnel. The final locality, geographic coverage, operating organization, accessible datasets, routing and allocation methods, and evaluation design remain subject to confirmation.

1.2 Purpose and Description

The purpose of this study is to develop *OverFlow*, a localized, scenario-based decision-support system for flood-evacuation planning. A planner configures rainfall intensity and duration, affected population by barangay, evacuation-center status, center capacity, manual road or bridge closures, and available resources. Using the geographic and terrain data for the selected study area, the system represents the resulting flood and evacuation conditions in an interactive two-dimensional planning environment.

For each scenario, the system identifies flood-affected areas and analyzes evacuation options under the configured constraints. It is intended to identify feasible and alternative routes, assign affected groups to available evacuation centers, estimate evacuation time, and present resource-allocation information. The planner can compare the results of different scenarios and use them as a reference in preparing or revising an evacuation plan.

The output is decision-support information, not an autonomous emergency decision or a guarantee of actual road safety. Results depend on the completeness, accuracy, resolution,

and currency of the underlying data, as well as on the selected scenario inputs. The simplified terrain-based flood computation is suitable for comparing relative conditions, but it does not model detailed hydraulic behavior, flow velocity, infiltration, or drainage pipes and canals.

1.3 Objectives of the Study

The general objective of this study is to develop and evaluate a localized two-dimensional simulation-based decision-support system for creating and comparing flood-evacuation planning scenarios.

Specifically, it aims to:

1. integrate validated local map, road, hazard, evacuation-center, population, and resource data for the selected study area;
2. represent flood-affected areas from DEM-based terrain accumulation using rainfall intensity and duration, together with road disruption, evacuation demand, center capacity, and resource availability;
3. analyze feasible evacuation routes, center allocation, estimated evacuation time, and resource distribution under configured scenario constraints;
4. present scenario results through an interactive two-dimensional map and comparison interface; and
5. evaluate the developed system using methods and metrics to be selected for the final study.

1.4 Significance of the Study

The study is intended to provide an interactive way of examining localized flood-evacuation scenarios. By showing flood-affected areas together with road accessibility, evacuation demand, center capacity, and resources, *OverFlow* may help users compare the consequences of alternative planning conditions before an emergency occurs.

For local disaster planners and responders, the system may provide a supplementary reference for preparing and revising evacuation plans. It can support the examination of feasible routes, alternate routes, center allocation, estimated evacuation time, and resource requirements under different scenario conditions. It does not replace existing disaster-management procedures or the judgment of authorized personnel.

For local engineering and planning personnel, barangay responders, and evacuation-center managers, the system may support coordination by presenting relevant geographic and capacity information in one planning environment. Community residents are intended indirect beneficiaries when better-informed preparedness and evacuation plans are developed. The study may also provide future researchers and developers with a reference for localized, terrain-based flood-evacuation decision-support systems.

1.5 Scope and Delimitations of the Study

This study covers the development of *OverFlow*, a localized two-dimensional flood-evacuation planning and decision-support system. It includes configurable scenario inputs for rainfall intensity and duration, affected population by barangay, evacuation-center status and capacity, manual road or bridge closures, and available rescue resources. The system uses DEM data to compute terrain-based flood-affected areas through D8 flow direction, flow accu-

mulation, rainfall overlay, and threshold-based classification. It represents roads, bridges, barangay boundaries, evacuation centers, population information, and resource conditions to support analysis and comparison of evacuation scenarios.

The system produces decision-support outputs that include flood-affected areas, recommended and alternative evacuation routes, center allocation, estimated evacuation time, resource-allocation information, dynamic maps, and scenario comparisons. Drainage infrastructure may be included only as a reference map layer. The final study area, deployment organization, and availability of local datasets remain subject to confirmation; the system's implemented coverage and features will be bounded by data that can be legitimately obtained and validated during the study.

The study excludes real-time flood forecasting, public navigation, emergency dispatch control, and direct control of physical infrastructure. It does not implement a calibrated hydraulic model and does not claim exact flood depth, velocity, or prediction of actual flood behavior. The model does not represent detailed hydraulic processes such as soil infiltration, drainage pipes and canals, or multi-directional flow spreading. Its outputs are intended for relative scenario comparison and human planning support, not as absolute emergency instructions.

1.6 Operational Definition of Terms

Affected Population. Refers to the number of people assigned to a barangay or area within a configured evacuation scenario.

D8 Flow Direction. Refers to a terrain-processing method that assigns each DEM cell a downhill flow direction toward its steepest neighboring cell among eight possible neighbors.

Decision-Support System. Refers to a system that helps a human decision-maker

evaluate options. In *OverFlow*, it presents processed information for examining flood-evacuation scenarios; it does not make autonomous emergency decisions.

Digital Elevation Model (DEM). Refers to a two-dimensional grid in which each cell stores ground elevation. It is the primary terrain input for the flood computation in *OverFlow*.

Drainage Infrastructure. Refers to pipes, canals, and other structures that collect or convey water. In *OverFlow*, available drainage information is a reference map layer and is not used to compute flooding.

Evacuation-Center Allocation. Refers to the assignment of affected groups to evacuation centers while considering accessibility and available capacity.

Evacuation Route. Refers to a road-network path considered usable between an affected area and an evacuation center under a scenario's constraints.

Flood Scenario. Refers to a configured set of conditions, including rainfall, affected population, road closures, center status and capacity, and available resources, used to examine a possible evacuation situation.

Flow Accumulation. Refers to a grid operation that counts or estimates the upstream area draining into a DEM cell, indicating locations where water may collect.

OverFlow. Refers to the proposed localized two-dimensional, simulation-based decision-support system for flood-evacuation planning developed in this study.

Terrain-Based Flood Computation. Refers to the simplified computation of flood-affected areas from rainfall input and DEM terrain data using D8 flow direction, flow accumulation, rainfall overlay, and threshold-based flood classification. It is not a hydraulic model.

2 Review of Related Literature

2.1 Introduction

This review examines the studies and systems that provide the main research and methodological foundations for a simulation-based decision support system for flood evacuation planning. The reviewed works address four connected concerns: the effect of flooding on road accessibility, the assignment of affected populations to evacuation centers, the simulation and comparison of evacuation scenarios, and the use of terrain and behavioral information in evacuation analysis. Recent studies from 2023 to 2025 are emphasized to reflect current geospatial and optimization approaches. Earlier works are retained where they establish a foundational method, document an influential related system, or provide locally relevant evidence that remains directly applicable to the study.

Taken together, the literature shows that flood evacuation planning is not adequately represented by a static map or a shortest route calculated under normal road conditions. A useful planning system must relate the flood condition to the road network, determine which destinations remain reachable, respect shelter capacity and other operational constraints, and allow planners to compare the consequences of different assumptions. These requirements correspond to the objectives of OverFlow: to integrate terrain, road, population, evacuation-center, and resource information; identify feasible routes and center allocations; and present alternative scenarios through an interactive geospatial interface.

2.2 Flood-Aware Road Accessibility and Geospatial Decision Support

Alabbad et al. developed a web-based decision support framework that evaluates road and bridge accessibility under multiple flood conditions and provides shortest-path, facility-allocation, and service-area analyses [1]. Its tests in Cedar Rapids and Charles City showed that inundation can disconnect parts of a road network, lengthen routes, and change which critical facility is nearest and reachable. The study is particularly relevant because it treats flood extent as an operational constraint on a graph rather than as a visual overlay alone. It also demonstrates that flood-related network analysis can be delivered through a web interface to users who do not operate specialist GIS software.

The framework supports OverFlow's proposed integration of flood-affected areas with road routing and its use of an accessible map-based interface. Its scope, however, is centered on transportation accessibility and emergency-facility placement. It does not provide a complete evacuation-planning model that jointly accounts for affected populations, shelter capacities, available resources, and estimated evacuation time. Its use of prepared inundation maps and two Iowa case areas also means that its results cannot be transferred directly to a Philippine locality without locally appropriate terrain, road, and population data. OverFlow can adopt the study's flood-constrained graph principle while evaluating it within a localized planning workflow.

Suwanno et al. combined GIS-based flood susceptibility analysis with road-network analysis to identify suitable evacuation areas and routes in Nakhon Si Thammarat, Thailand [2]. Elevation, slope, land use, proximity to water, and other spatial criteria were used to locate flood-prone areas and assess evacuation alternatives under normal and flooded conditions. Because the case is situated in Southeast Asia, it provides a closer geographic

and environmental comparison than studies based only in North America. It also confirms that the safety of a destination and the accessibility of its connecting route must be evaluated together.

The study offers strong support for using GIS layers to screen evacuation centers and calculate routes, but its result is principally a set of suitable areas and paths derived from the selected criteria and weights. Such an approach is valuable for preparedness, yet it provides limited support for repeated what-if comparisons involving road closures, changing demand, center status, and resource availability. For OverFlow, the study informs the selection and spatial validation of routes and centers, while the proposed scenario mechanism extends the analysis by allowing planners to change operational conditions and compare the resulting plans.

2.3 Shelter Capacity and Population Allocation

Sharbaf et al. proposed a risk-based shelter-network design model that integrates population exposure, shelter risk, evacuation risk, pedestrian accessibility, and flood conditions in a single optimization framework [3]. Applied in Haiti, the model seeks to maximize coverage of vulnerable populations while reducing the risks associated with shelter locations and evacuation paths. Its contribution is the recognition that a nearby shelter is not automatically an appropriate assignment: both the facility and the route leading to it may be exposed to flooding. The study therefore provides a current basis for treating evacuation-center allocation as a constrained risk problem rather than a nearest-facility lookup.

This model directly supports OverFlow's objective of assigning affected populations to accessible centers while respecting safety and capacity considerations. Nevertheless, Sharbaf et al. focus on strategic shelter-network design and pedestrian evacuation, whereas

OverFlow is intended to help local planners test operational scenarios involving existing centers, road interruptions, and available resources. The difference suggests that OverFlow should not claim to reproduce the study's optimization model unless the same objective function, constraints, and validation procedures are implemented. Instead, the study supplies criteria against which OverFlow's allocation method can be defined and evaluated, including capacity satisfaction, population coverage, and route exposure.

Gama et al. formulated a multi-period shelter location-allocation model that determines when shelters should open, when evacuation orders should be issued, and how evacuees should be assigned as flood and road conditions change over time [4]. The model accounts for limited shelter capacities, time-dependent travel conditions, and the gradual availability of facilities. It is older than the preferred five-year review period, but it remains important because it connects the temporal development of a flood with capacity-constrained allocation and demonstrates why a single static assignment may be insufficient.

The authors also report that exact optimization becomes computationally demanding for realistic problem sizes and therefore use a simulated-annealing heuristic. This limitation is relevant to the design of OverFlow: a planning tool must return understandable results within a practical interaction time, even when an exact optimum is expensive to compute. The study supports scenario comparison and time-sensitive allocation, while also indicating that OverFlow must define its problem size, objective function, and acceptable solution time before describing any route or allocation as optimal.

2.4 Evacuation Simulation and Related Systems

Aketa et al. developed a GIS-based evacuation action analysis system that links flood analysis with network analysis for multiple evacuees and shelters [5]. The system simulates

evacuation movement, considers walking speed, and searches routes across a road network to support route and shelter planning. Although published well outside the recent review window, it is retained as an early related system that demonstrates the feasibility of coupling flood conditions with network-based evacuation behavior in one geospatial environment.

Its age and brief reported evaluation limit its value as evidence for current web architecture or modern local-government deployment. More significantly, the system emphasizes route and shelter analysis but does not establish the broader scenario controls, resource-allocation outputs, or comparative planning interface proposed for OverFlow. Its relevance is therefore conceptual: it shows that flood and evacuation analyses should interact, while the present study must establish a current implementation and evaluate whether planners can use its combined outputs effectively.

The United States Army Corps of Engineers' LifeSim 2.0 is a more comprehensive related system for simulating population redistribution, warnings, evacuation traffic, flood exposure, and potential life loss [6]. Its agent-based model represents interactions among evacuees, roads, vehicles, and an advancing hazard; it can reroute groups when roads flood and uses repeated simulations to represent uncertainty. LifeSim demonstrates the analytical value of treating evacuation as a dynamic process and of comparing alternative warning, road, and hazard scenarios rather than relying on a single deterministic result.

LifeSim also clarifies the intended boundary of OverFlow. Reproducing an agent-based traffic and life-loss model would require behavioral, vehicle, warning, hazard-intensity, and validation data beyond the confirmed scope of the project. OverFlow can apply the more limited but relevant principle of configurable scenario comparison without presenting its terrain-based accumulation output as a calibrated hydraulic forecast or its planning estimates as life-loss predictions. LifeSim therefore serves both as a design reference and as evidence

for clearly documenting model assumptions and uncertainty.

2.5 Terrain Processing and Evacuation Behavior

O’Callaghan and Mark introduced the method now commonly called D8, in which each cell in a digital elevation model directs flow toward its steepest downslope neighbor, allowing drainage direction and accumulated contributing area to be derived from gridded terrain [7]. This is a classic methodological source rather than a recent evacuation study, and it provides the foundation for OverFlow’s confirmed terrain-based accumulation approach. D8 offers a computationally straightforward way to identify likely flow paths and areas of accumulation from elevation data.

The method does not by itself calculate flood depth, velocity, drainage-system behavior, or the timing of inundation. Consequently, terrain-derived accumulation should be used in OverFlow to classify scenario-specific flood-affected areas according to an explicitly documented threshold and should not be described as a calibrated hydraulic model. This distinction affects the methodology: the study must validate the digital elevation model, document preprocessing and threshold choices, and test how the resulting flood classification changes the set of usable road links.

Lim et al. examined the route choices of households affected by flooding in Bagong Silangan, Quezon City, using three binary logit models based on hazard, socioeconomic, and evacuation-related variables [8]. Their Philippine case shows that evacuees do not necessarily select routes solely by minimum distance; departure timing, familiarity, perceived conditions, and household characteristics can influence movement. The study provides local evidence that a mathematically shortest route and an actual human route are not always equivalent.

This finding is critical for interpreting OverFlow's outputs. A feasible or recommended path should be presented as a planning option derived from the available network and scenario constraints, not as a prediction of the route every resident will follow. The study also identifies a methodological opportunity for evaluation: proposed routes can be reviewed by local disaster personnel and, where data are available, compared with known routes, accessibility conditions, and community practices. Behavioral modeling may remain outside the initial implementation, but its absence should be stated as a limitation when evacuation-time estimates are interpreted.

2.6 Synthesis and Research Gap

3 Methodology

3.1 Research Design

3.2 Data Gathering Tools

3.3 Research Instruments

3.4 Data Gathering Procedures

3.5 Statistical Treatment of Data

4 Results and Discussion

4.1 System Development and Scenario-Analysis Results

4.2 System Evaluation Results

5 Conclusions and Recommendations

5.1 Conclusion

5.2 Recommendation

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APPENDICES

A Detailed Tables

B Questionnaires